DEVSECOPS LAB BOOK · PHASE 1: FOUNDATION

Chapter 02

BIND9: Split-Brain DNS Deep Dive

Building an authoritative, view-based DNS server that hides internal topology from the outside world — and becomes the foundation every other lab service depends on.

`named · views · zones · ACLs · dig`

2.1 — THE PROBLEM

Why DNS Exists

The Address Book Analogy

You don't memorize Sarah's number *(+1-415-555-0142)* — you search her name in your contacts.

DNS is the internet's address book. A browser visiting gitlab.devsecops.com has no idea what that means — the network only understands IPs like 10.0.2.14. DNS translates the human name into the machine number.

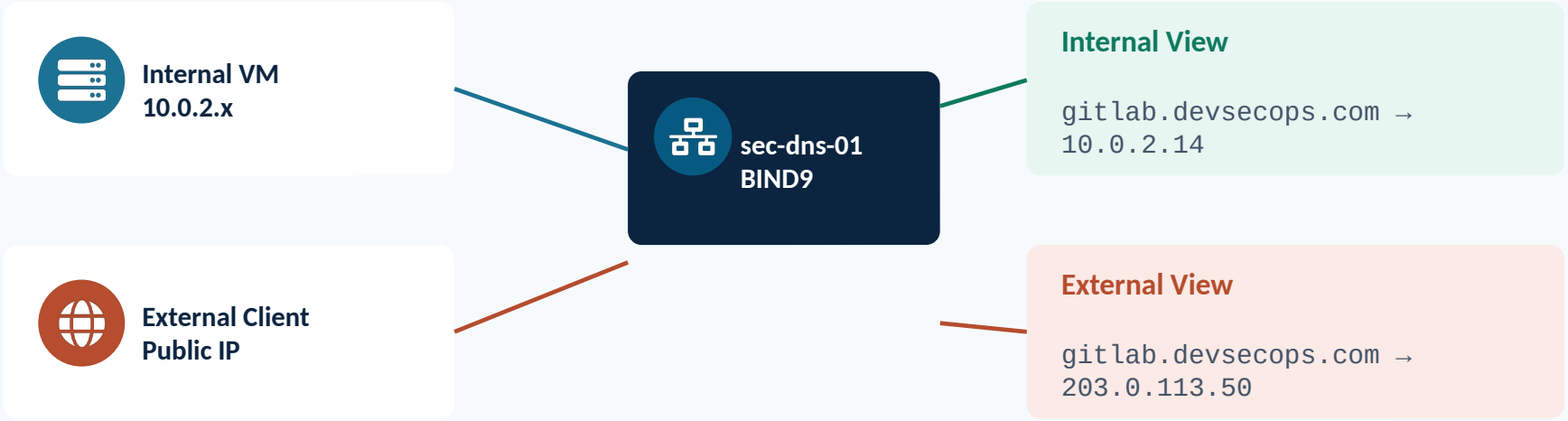
What Breaks Without It

- Hostname-based SSH configs fail
- TLS certs can't be issued (need DNS names)
- Authentik OIDC redirects need a domain
- Kubernetes service discovery depends on DNS
- LDAP can't use FQDN-based binds
- Log correlation breaks — logs show raw IPs

2.2 — CONCEPT

What Is Split-Brain DNS?

Split-brain DNS (split-horizon) is a configuration where the same domain name resolves to a different IP address depending on who is asking.



Security benefit: external clients never see your internal IP scheme — only what you deliberately expose.

2.3 — ARCHITECTURE

BIND9 Process Overview (named)

named.conf — master

options — global

local — zones/views

logging — log config

VIEWS (split-brain)

internal view

`ACL: 10.0.2.0/24`

external view

`ACL: any`

ZONE FILES

`db.devsecops.com.internal`

Forward · name → IP (internal)

`db.devsecops.com.external`

Forward · name → IP (external)

`db.10.0.2`

Reverse · IP → name

dig / systemd-resolved → UDP/TCP port 53

2.3 — VOCABULARY

Key BIND9 Concepts

Concept	Definition	Lab Usage
Zone	Portion of DNS namespace managed by this server	devsecops.com zone
Forward Zone	Maps names → IP addresses (A records)	gitlab.devsecops.com → 10.0.2.14
Reverse Zone	Maps IP addresses → names (PTR records)	10.0.2.14 → sec-vcs-01.devsecops.com
View	A named set of zones served to a specific client ACL	internal vs external
ACL	Access Control List — who can use which view	10.0.2.0/24 = internal clients
SOA Record	Start of Authority — zone metadata (serial, refresh)	Required in every zone
NS Record	Declares who is authoritative for the zone	Points to sec-dns-01
CNAME	Canonical Name — alias from one name to another	gitlab → sec-vcs-01
PTR	Pointer — maps an IP to a name (reverse lookup)	Used by SSH, logging, SIEM

2.4 — SETUP

Installation & Verification

```
$ sudo apt update
$ sudo apt install -y bind9 \
    bind9utils bind9-doc dnsutils
```

Package	Purpose
bind9	The BIND9 daemon (named) — the DNS server itself
bind9utils	named-checkconf, named-checkzone validation tools
bind9-doc	Offline documentation for record types
dnsutils	dig, nslookup, host — query tools for testing

Verify the Service

```
named -v
```

→ *BIND 9.18.x-Ubuntu (ESV)*

```
sudo systemctl status bind9
```

→ *Active: active (running)*

```
sudo ss -tulnp | grep named
```

→ *udp/tcp LISTEN on 0.0.0.0:53*

BIND9 must bind BOTH UDP and TCP on port 53.

2.5 — LAYOUT

Configuration File Structure

```
/etc/bind/
├── named.conf
├── named.conf.options
├── named.conf.local
├── named.conf.logging
└── zones/
    ├── db.devsecops.com.internal
    ├── db.devsecops.com.external
    └── db.10.0.2
```

`named.conf`

Master config — only includes the others

`named.conf.options`

Global behaviour: recursion, forwarders, rate limits

`named.conf.local`

Our ACLs, views, and zone definitions

`named.conf.logging`

Channels + categories for logging

`zones/`

The actual zone data files (forward + reverse)

```
Setup: sudo mkdir -p /etc/bind/zones  &&  sudo chown -R bind:bind /etc/bind/zones
```

2.5 — NAMED.CONF.OPTIONS

Global Options: Security-First Defaults

`forward only + forwarders`

Never uses root hints; always forwards to 1.1.1.1 / 8.8.8.8

`allow-transfer { none; }`

No one gets a full zone copy (AXFR) by default

`dnssec-validation auto`

Validates DNSSEC signatures — protects against spoofing/poisoning

`rate-limit 10/5s`

Throttles responses per client — blunts amplification attacks

`listen-on 127.0.0.1, 10.0.2.10`

Binds only to loopback + the lab interface, not 0.0.0.0

`max-cache-size 64m`

Bounds memory used by the resolver cache

`allow-recursion { localnets; }`

CRITICAL: never allow recursion from the public internet

`version "DNS Server"`

Hides the real BIND version from reconnaissance queries

2.5 — NAMED.CONF.LOGGING

Logging: Channels → Categories

Channels (destinations)

`bind_log`

/var/log/named/named.log — 5×20MB, dynamic severity

`security_log`

/var/log/named/security.log — ACL violations, TSIG

`syslog_channel`

→ rsyslog → Elastic Stack, severity info

Categories (event types) → routed to channels

- queries — every DNS query
- security — ACL violations, TSIG
- config / dnssec / xfer-in / xfer-out
- update / notify / resolver
- lame-servers → null (suppressed noise)

Why it matters: good logging turns a 10-hour debugging session into a 10-minute one — and every query becomes a security-observable event.

2.5 — ZONE FILES

Internal vs. External Forward Zones

db.devsecops.com.internal

- TTL 300s — fast propagation for lab changes
- 12 A records: every VM by real internal IP
- 20+ CNAMEs: gitlab, vault, auth, harbor, kibana…
- Wildcard * → sec-edge-01 (dynamic app routing)
- Served only to internal_clients ACL

db.devsecops.com.external

- TTL 3600s — slower change cadence expected
- Zone apex + gitlab/harbor/vault only
- All point to sec-edge-01 (10.0.2.12) in the lab
- NO internal hostnames, NO infra IPs exposed
- Served to match-clients { any; }, no recursion

2.5 — REVERSE ZONE

PTR Records: IP → Hostname

Reverse DNS maps an IP address back to a hostname. The zone name reads backwards because DNS resolves right-to-left: 2.0.10.in-addr.arpa covers all of 10.0.2.0/24.

- SSH connection logging
- SIEM log enrichment
- Authentication systems
- Email anti-spam (SPF/DKIM)

```
10   IN  PTR sec-dns-01.devsecops.com.     ; 10.0.2.10
14   IN  PTR sec-vcs-01.devsecops.com.      ; 10.0.2.14
20   IN  PTR k8s-master-01.devsecops.com.   ; 10.0.2.20
30   IN  PTR sec-obs-01.devsecops.com.      ; 10.0.2.30
```

PTR format: <last-octet> IN PTR <fqdn>. (the zone provides the rest of the address)

2.5 — NAMED.CONF.LOCAL

Views & ACLs — The Split-Brain Engine

ACL

```
acl "internal_clients" {
  10.0.2.0/24;
  127.0.0.1;
  ::1;
};
```

view "internal"

- match-clients { internal_clients; }
- recursion yes (resolve external names)
- zone devsecops.com → type master
- zone 2.0.10.in-addr.arpa → master
- zone "." → hint (root servers)
- localhost / loopback zones

view "external"

- match-clients { any; }
- recursion no (blocks open-resolver abuse)
- zone devsecops.com → master (public-safe file)
- no reverse zone, no root hints

2.6 – 2.7

Validate, Start & Enable BIND9

1	`named-checkconf`	Parses all included files. Silence = success. Errors show file:line.
2	`named-checkzone <zone> <file>`	Run for internal, external, and reverse zones. Expect "OK".
3	`sudo chown -R root:bind /etc/bind`	chmod 640 on configs/zones so the bind user can read them.
4	`sudo systemctl restart bind9`	Restart (not reload) after structural changes like new views.
5	`sudo systemctl enable bind9`	Ensures BIND9 comes back up automatically after reboot.

💡 *Unlike most Unix tools, named-checkconf's silence means everything is correct.*

2.8 – 2.9

Firewall & Testing DNS Resolution

Firewall (ufw)

```
sudo ufw allow from 10.0.2.0/24 \
  to any port 53 proto tcp
sudo ufw allow from 10.0.2.0/24 \
  to any port 53 proto udp
```

If +norecurse from outside returns real answers, your resolver is open — usable for DNS amplification DDoS. Tighten allow-recursion immediately.

Forward lookup

`dig @10.0.2.10 sec-vcs-01.devsecops.com A → 10.0.2.14`

CNAME chain

`dig gitlab.devsecops.com → CNAME sec-vcs-01 → A 10.0.2.14`

Reverse (PTR)

`dig -x 10.0.2.14 → sec-vcs-01.devsecops.com.`

External view

`+subnet=203.0.113.1/32 → returns edge IP 10.0.2.12`

Recursion works

`dig ubuntu.com A → real public IPs returned`

No open resolver

`+norecurse from outside → REFUSED (must fail!)`

2.10 — ROLLOUT

Pointing the Lab at sec-dns-01

Netplan — every non-DNS VM

```
nameservers:
  addresses:
    - 10.0.2.10
  search:
    - devsecops.com

sudo netplan apply
```

Remove the 8.8.8.8 fallback once BIND9 is confirmed handling external queries.

Verify from Another VM

```
resolvectl status
```

```
→ DNS Servers: 10.0.2.10
→ Search Domains: devsecops.com
```

```
dig vault.devsecops.com
```

```
→ vault.devsecops.com. CNAME sec-vault-01.
→ sec-vault-01.devsecops.com. A 10.0.2.13
```

Every VM in the lab is now reachable by name — not fragile, memorized IPs.

2.11 — OPERATIONS

Making Zone Updates (Change Process)

Edit

```
sudo vim db.devsecops.com.internal
```

Bump serial

```
2026010101 → 2026010102
```

Validate

```
named-checkzone <zone> <file>
```

Reload

```
sudo rndc reload  (no downtime)
```

Verify

```
dig @10.0.2.10 <new-record>
```

Forget the serial bump and BIND9 considers the zone unchanged — secondaries (and rndc reload) won't pick up your edit.

Serial Convention — YYYYMMDDnn

```
2026010101 — Jan 1, 2026, 1st change
```

```
2026010102 — Jan 1, 2026, 2nd change
```

```
2026010201 — Jan 2, 2026, 1st change
```

2.12 – 2.13

Automated Validation & Troubleshooting

validate-chapter-02.sh

- BIND9 service is active
- 10 infrastructure A records resolve
- 6+ service CNAMEs resolve correctly
- Reverse PTR lookups match hostnames
- External forwarding reaches ubuntu.com
- Pass/Fail counters, exit 1 on failure

Common Failures

"unknown option"

Typo — read file:line in the checkconf error

Queries time out

Wrong listen-on interface, or ufw blocking :53

Internal gets external results

match-clients ACL doesn't include client's subnet

Reload doesn't apply

Serial number wasn't incremented

dig returns SERVFAIL

Check /var/log/named/named.log for the root cause

2.14 — SUMMARY

Chapter Summary & Key Takeaways

What We Built	Why It Matters
Split-brain DNS with two views	Internal/external resolution without leaking internal IPs
All 12 VMs registered by hostname	Every service reachable by name, not fragile IP
15+ service CNAMEs	Abstraction layer — move a service, update one CNAME
Reverse DNS (PTR records)	SSH logs, SIEM enrichment, auth systems all need it
Query logging enabled	Security-critical: every DNS query is an observable event
Recursion locked to internal	Cannot be abused as an open resolver

DNS is your first security control: a VM querying malware-c2.attacker.com at 3 AM shows up in DNS logs before the connection completes.